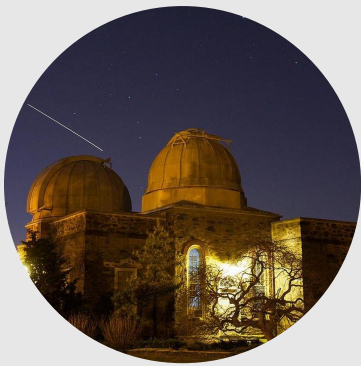




Observational Astronomy

ASTR 341

Instructor Info —



Prof. Karen Masters



Student Hrs: Zoom drop-in time TBD; book 9-5 at: calendly.com/karenlmasters; dark sky hours by email/Slack



Observatory A or online



www.haverford.edu/users/klmasters



klmasters@haverford.edu

Course Info —



Prereq: ASTR204



Mon, 7.30-10pm



Observatory Classroom

Overview

Astro 341 consists primarily of observing projects using the CCD camera on the 16" Schmidt-Cassegrain telescope (in the large dome of the Strawbridge Observatory), techniques of data mining from astronomical catalogues and/or using small radio telescope(s) during an (optional) weekend trip to the Green Bank Observatory. Data from your observations will be analyzed with either an image processing software package (AstroImageJ) or Python. Two sets of results will be written up as formal reports in the style of scientific papers and one will be presented at the end of the semester. Various scaffolding assignments across five themes related to observational astrophysics will be done by all students, used to support project work. For a visual overview of the themes, assignments and projects see the last page.

The course will be informal/student led. After the initial instructional observing sessions, each observing team (consisting of three students) will have the responsibility for scheduling observing with the 16" telescope. The regularly scheduled "class time" (Mondays, 7:30-10:00 p.m.) will serve three purposes. First, some of this time will be used for discussion on a variety of topics; e.g., an introduction to image processing, operating the 16" Schmidt-Cassegrain telescope, an introduction to CCD cameras and the data they create, introduction to radio astronomy. In addition, we will meet during this time to discuss the details of individual projects and reports. Finally, this time slot makes sure that there is at least one night a week when members of all observing teams do not have scheduling conflicts. On clear nights one team will observe. Because observing sessions may need to last for 5 or 6 hours, do not make any other commitments on Monday evenings that you cannot reschedule. Because of the weather, there may be other days of the week when you might have to observe. Plan to observe the first night it is clear that you are ready. Don't waste clear nights - they are a precious commodity for (optical) observational astronomers.

Material

Main Text:

An Introduction to Observational Astrophysics, by Mark Galloway

Supplementary:

Handbook of CCD Astronomy, by Steve Howell

Open Stax Astronomer2e, free online

Essential Radio Astronomy, by James J. Condon and Scott M. Ransom (free eBook).

List of Topics/Projects

All students will do all seven assignments - collectively worth 30% of total grade. The weakest grade will be dropped.

Every student must submit two projects (worth 30% each). If a student chooses to do more than two the best two will be counted.

Resubmissions will be permitted for any assignment or project below 80%.

Class participation including effort in observing (via logs), worth 10% of final grade.

Telescope Basics	Telescope basics – pass 16" checkout – three-filter imaging of a Messier Object – [Messier Marathon Project]
Radio Astronomy	Sources of radio emissions – either assignment on radio telescopes OR GBO trip diary – [Radio Astronomy Project based on GBO trip]
Optics/Instruments	Assignment on CCDs – [Project characterizing the CCD camera]
Science Obs	Read and report on observational paper – Assignment on exoplanets – [Project to Observe Exoplanet Transits with the 16"]
Data Mining	Assignment based on Gaia HR diagram – [Project to observe star cluster and match to Gaia]

FAQs

? Will I learn how to use a telescope in this class?

! Yes. That's the point - but we'll do more than just use the telescope - we'll use them to learn about astrophysics!

? What is astrophysics?

! The use of physics to understand and learn about objects in the night sky. All areas of physics, as well as a lot of chemistry (and some biology) are important to the full understanding of astrophysical objects.

? What's the difference between astronomy and astrophysics?

! It's really just semantics in the modern usage. The most useful distinction is that there are lots of Amateur Astronomers (someone who as a hobby uses a telescope to view the skies), while it's much more unusual to be an Amateur Astrophysicist (someone who uses physics to interpret objects in the Universe). At Haverford the Astrophysics Major is basically the same as the Physics Major with an Astro emphasis, while the Astronomy Major has more astronomy and less core physics.

? What is your favourite astronomical object?

! Galaxies. Which are challenging to observe from near to Philly, but not impossible.... I've done the Whirlpool galaxy (M51). Other bright nearby galaxies are possible.

Honour Code

Collaboration is an important part of science. You are strongly encouraged to work together and/or consult one another for work in this class. You must do observing in groups of three. You are encouraged to consult any books necessary as well as resources on the internet. You must, however, turn in your own individual assignments/project reports, and this must be written on your own. Copying and pasting (even parts of sentences) is not permitted and is a violation of the Honour Code. Good collaboration involves everyone understanding what is going on. Therefore even if the basic solution is shared you must explain it in your own words (including mathematical words). Please list any students that you collaborated with. Please pay attention to your classmates to make sure no one is being left out of collaborative work.

Diversity and Inclusivity Statement

Our classroom should be a place where all members will be treated with respect. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. If something was said in class (by anyone including myself) that made you feel uncomfortable, please talk to me about it (anonymous feedback is always an option). I appreciate any opportunity to continue my learning about diverse perspectives.

In an ideal world, science would be objective. However, science is done by people, and is historically built on a small subset of privileged voices. In this class, we will make an effort to read work from a diverse group of scientists, but limits still exist on this diversity. I believe that integrating a diverse set of experiences is important for a more comprehensive understanding of science. We may discuss issues of diversity in astrophysics as part of the course from time to time. Please contact me (in person or electronically) or submit anonymous feedback if you have any suggestions to improve the quality of the course materials.

Accommodation Statement

Haverford College is committed to providing equal access to students with a disability. If you have (or think you have) a learning difference or disability - including mental health, medical, or physical impairment, please contact the Office of Access and Disability Services (ADS) at hc-ads@haverford.edu. The Coordinator will confidentially discuss the process to establish reasonable accommodations.

Students who have already been approved to receive academic accommodations and want to use their accommodations in this course should share their verification letter with me and also make arrangements to meet with me as soon as possible to discuss their the specific accommodations. Please note that accommodations are not retroactive and require advance notice to implement.

It is a state law in Pennsylvania that individuals must be given advance notice if they are to be recorded. Therefore, any student who has a disability-related need to audio record this class must first be approved for this accommodation from the Coordinator of Access and Disability Services and then must speak with me. Other class members will need to be aware that this class may be recorded.

Draft Class Schedule

Date	Topic/Plan/Activities	Deadlines (Draft)
Telescope Basics, Optics and Instruments		
Week 1 Aug 29th	Introductory material (telescopes, co-ordinates, magnitudes etc)	Assignment on Intro material and planning 3-colour image obs due Friday
Week 2 Sept 5th	Data intro pretty pictures/AstroImageJ working time in class How to make a 3-colour image using a CCD; physics of CCD cameras. Advice on Messier Marathon and/or CCD projects	Assignment on CCDs due Friday
Radio Astronomy		
Week 3 Sept 12th	Introduction to radio astronomy - Working time on 3-colour image.	Assignment on Radio emissions due Friday.
Week 4 Sept 19th	More radio astronomy content. Discuss project choices	Nothing due
Fri 23rd-Sun 25th	Field Trip to Green Bank Observatory (optional)	
Week 5 Sept 26th	Rosh Hashana. NO CLASS if cloudy – observing help if clear	Assignment on radio telescopes and techniques due Friday OR diary of observing trip due Fri- day.
Week 6 Oct 3rd	Working time	Three-filter image assignment due Friday.
Oct 8-16	FALL BREAK	
Science Observing and Data Mining		
Week 7 Oct 17th	Progress report on first project in class Exoplanet transits content. Advice on Exoplanet transit project.	Exoplanet transit assignment due Friday.
Week 8 Oct 24th	Stars, HR diagrams and Gaia. Advice on Star cluster observing project	Gaia data HR diagram due Friday.
Week 9 Oct 31st	Selection of observational paper . Discussion/planning time for talks	Nothing due
Week 10 Nov 7th	Progress report on projects in class. Working time. Cloudy activity: Cultural significance of astronomy	Nothing due
Week 11 Nov 14th	Talks on observational papers (reschedule if clear)	Nothing due
Week 12 Nov 21st	Thanksgiving week.	Nothing due

Progress report on projects in class. Working time.

Cloudy activity: Cultural significance of astronomy

Week 13 Nov 28th	(Monday after Thanksgiving)	1st Project due Friday 2nd December.
------------------	-----------------------------	--------------------------------------

Drop in help (optional)

Week 14 Dec 5th	Talks on 1st project.	Nothing due
-----------------	-----------------------	-------------

Progress report on 2nd project in class.

Finals (12-16th Dec)

End of finals	Project 2 due Friday 16th December
---------------	------------------------------------

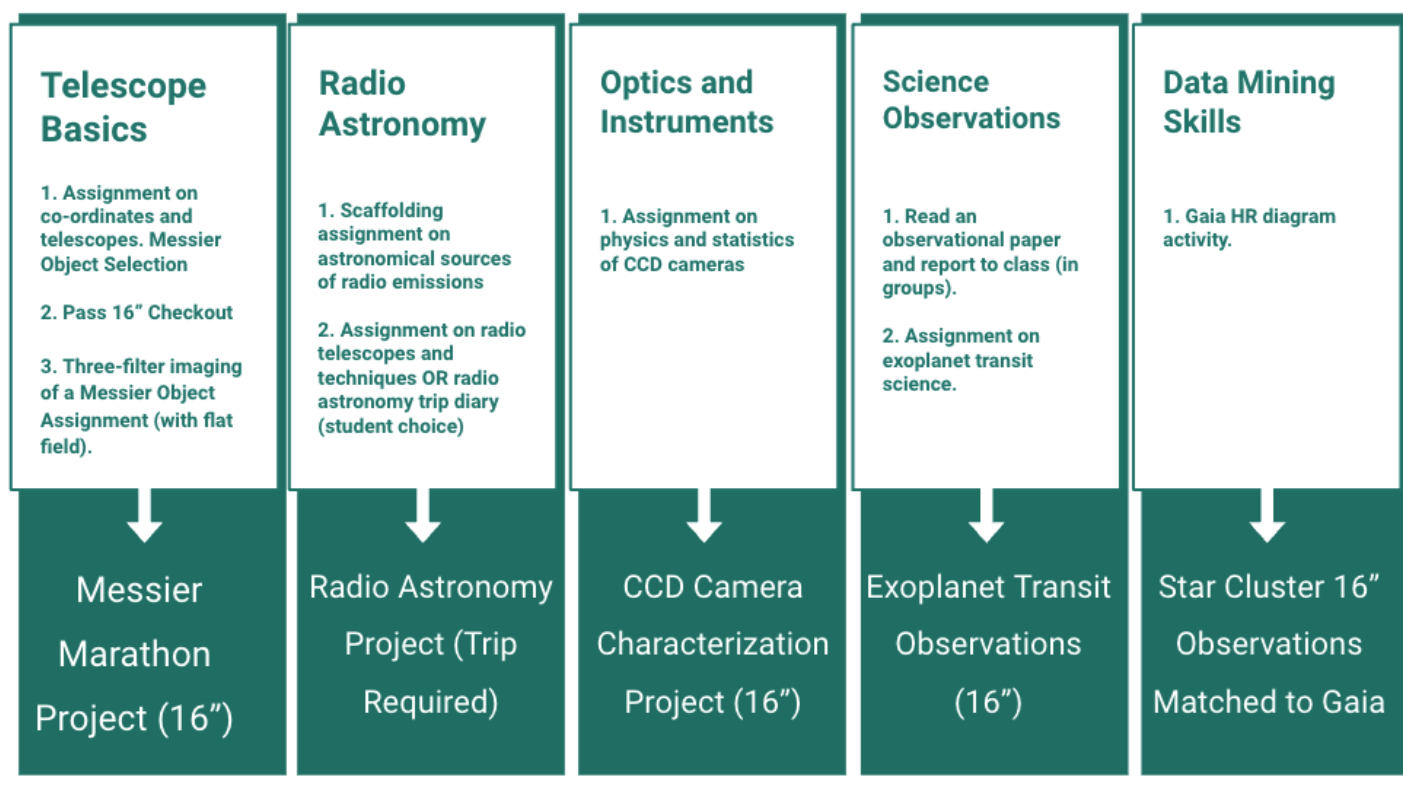


Figure 1: A graphical overview of assignments and projects for ASTR341. You do all assignments, and must do two projects.